Math 609 – Homework 5

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Problem 1

(a) Newton's method for roots

Applying Newton's method to solve

$$x^2 = a$$

gives the square-root iteration

$$t_{n+1} = \frac{1}{2}\left(t_n + \frac{a}{t_n}\right).$$

Answer:

Derivation of the square root formula

In order to get Newton's method to compute $\sqrt{a}$, we need to transform the function $x^2 = a$ so that $x = \sqrt{a}$. when $f(x) = 0$. The function with the necessary transformation is

$$f(x) = x^2 - a$$

The derivative of this funtion is

$$f'(x) = 2x$$

and Newton's method gives

$$t_{n+1} = t_n - \frac{f(t_n)}{f'(t_n)} = t_n - \frac{t_n^2 - a}{2t_n} = t_n - \frac{\cancel{t_n^2}}{\cancel{2t_n}} + \frac{a}{2t_n} = \frac{t_n}{2} + \frac{a}{2t_n} = \frac{1}{2}(t_n + \frac{a}{t_n})$$

Cubed quartic root derivation

Following the derivation of Newton's method for the square root, the same can be done to calculate the cubed root. First, transform x^3 to a function, where $x = \sqrt[3]{a}$ when $f(x) = 0$. The transformed function and following derivation is

$$f(x) = x^3 - a$$

$$f'(x) = 3x^2$$

$$t_{n+1} = t_n - \frac{t_n^3 - a}{3t_n^2} = t_n - \frac{\cancel{t_n^3}}{\cancel{3t_n^2}} + \frac{a}{3t_n^2} = \frac{2t_n}{3} + \frac{a}{3t_n^2} = \frac{1}{3}(2t_n + \frac{a}{t_n^2})$$

Using the prior derivations, the algorithm/formula to get the subsequent roots is

$$\frac{1}{root}((root-1)t_n + \frac{a}{t_n^{root-1}})$$

For instance, the quartic root is

$$t_{n+1} = t_n - \frac{t_n^4 - a}{4t_n^3} = \frac{3t_n}{4} + \frac{a}{4t_n^3} = \frac{1}{4}(3t_n + \frac{a}{t_n^3})$$

a=5

To get the squared root of 5, the initial guess, t_0 needs to be sufficietly close to the root of $f(x)$. For this case, $t_0 = 2$ would be the best choice, as it is the closest square root to $\sqrt{5}$.

After 4 iterations, the approximate squared root of 5 is 2.236.

(b) Fixed-point iteration

Consider

$$f(x) = x^3 - x + 1.$$

Find an appropriate interval and a function g such that the fixed-point iteration

$$x_{n+1} = g(x_n)$$

generates a sequence converging to a root of f.

Answer:

Finding the interval

To get the proper interval, a chosen function $g(x)$ needs to map back to the interval, $g([a,b]) \subseteq [a,b]$. If we choose a function $g(x_n) = \sqrt[3]{x-1}$, a good interval to start is $[-2,0]$, as $g(-2) = \sqrt[3]{-3}$ and $g(0) = -1$, thus the function maps into itself.

Show $g(x_n)$ is a contraction

In order to show that $g(x)$ converges on the interval $[-2,0]$ the function needs to be a contraction map. This is shown by

$$|g(x) - g(y)| \leq \lambda |x - y|$$

where $\lambda < 1$ and positive. Using the Mean Value Theorem, λ is the absolute value of the derivative of a point, inside the interval, if the absolute value of the derivative is less than one.

$$g'(x) = \frac{1}{3}(x-1)^{-2/3}$$

and

$$\lambda = |g'(0)| = |\frac{1}{3\sqrt[3]{4}}| < 1$$

Because $g(x)$ maps to itself and is a contraction on the interval, the iteration method will produce a sequence of solutions that converge. After 13 iterations, with an initial guess of 0 and stopping criteria of 1×10^{-9}, the approximate solution is -1.325.

Problem 2

Let α be a solution of

$$f(x) = 0,$$

and let $\{t_n\}$ be the sequence generated by Newton's method:

$$t_{n+1} = t_n - \frac{f(t_n)}{f'(t_n)}.$$

Show that

$$\alpha - t_{n+1} = -\frac{1}{2}\frac{f''(\xi_n)}{f'(t_n)}(\alpha - t_n)^2,$$

where ξ_n lies between t_n and α.

Answer:

By using the taylor series expansion, the function $f(t_n)$ with α as the center, which gives

$$f(\alpha) = f(t_n) + f'(t_n)(\alpha - t_n) + \frac{f''(\xi_n)}{2}(\alpha - t_n)^2$$

$$0 = f(t_n) + f'(t_n)(\alpha - t_n) + \frac{f''(\xi_n)}{2}(\alpha - t_n)^2$$

$$f(t_n) = -f'(t_n)(\alpha - t_n) - \frac{f''(\xi_n)}{2}(\alpha - t_n)^2$$

Newton's iteration can then be expanded

$$t_{n+1} = t_n - \frac{1}{f'(t_n)}(-f'(t_n)(\alpha - t_n) - \frac{f''(\xi_n)}{2}(\alpha - t_n)^2) = t_n + \frac{f'(t_n)}{f'(t_n)}(\alpha - t_n) + \frac{f''(\xi)}{2f'(t_n)}(\alpha - t_n)^2$$

$$t_{n+1} = t_n + (\alpha - t_n) + \frac{f''(\xi)}{f'(2t_n)}(\alpha - t_n)^2$$

$$t_{n+1} - \alpha = \frac{1}{2}\frac{f''(\xi)}{f'(t_n)}(\alpha - t_n)^2$$

$$\alpha - t_{n+1} = -\frac{1}{2}\frac{f''(\xi)}{f'(t_n)}(\alpha - t_n)^2$$

Problem 3

The nonlinear system

$$x_1^2 - 10x_1 + x_2^2 + 8 = 0,$$
$$x_1 x_2^2 + x_1 - 10x_2 + 8 = 0$$

can be transformed into the fixed-point system

$$x_1 = g_1(x_1, x_2) = \frac{x_1^2 + x_2^2 + 8}{10},$$

$$x_2 = g_2(x_1, x_2) = \frac{x_1 x_2^2 + x_1 + 8}{10}.$$

(a) show that $G = (g_1, g_2)^T$ has a unique fixed point

The interval for x_1, x_2 is $[0, 1.5]$. First, G needs to map to itself, $G : [0, 1.5] \times [0, 1.5] \subseteq [0, 1.5] \times [0, 1.5]$.

$$g_1(0,0) = \frac{4}{5}$$
$$g_1(1.5, 1.5) = \frac{2.25 + 2.25 + 8}{10} = \frac{12.5}{10} = 1.25$$
$$g_2(0,0) = \frac{4}{5}$$
$$g_2(1.5, 1.5) = \frac{3.375 + 1.5 + 8}{10} = \frac{12.875}{10} = 1.2875$$

From evaluating the functions at the ends of the interval, it is evident that the function maps the interval back into itself. Next, the function G needs to also be a contraction. The jacobian of G is

$$J_G = \begin{pmatrix} \frac{x_1}{5} & \frac{x_2}{5} \\ \frac{x_2^2+1}{10} & \frac{x_1 x_2}{5} \end{pmatrix}$$

In order for G to be a contraction, the maximum of $\|J_G\|_\infty$ needs to be less than one. Because of the increasing monoticity of the function, the maximum is at $x_1 = 1.5$ and $x_2 = 1.5$.

$$J_G(1.5, 1.5) = \begin{pmatrix} 0.3 & 0.3 \\ 0.325 & 0.45 \end{pmatrix}$$
$$\|J_G\|_\infty = \max(|0.3 + 0.3|, |0.325 + 0.45|) = 0.775$$
$$0.775 < 1$$

Because $\|J_G\|_\infty < 1$, G is a contraction. Due to the Banach fixed point theorem, G has a single fixed point.

(b) Result from code.

After running for 22 iterations with a starting point of $(1.5, 1.5)^T$ and a stopping tolerance for 1×10^{-9}, the result of the Fixed-Point iteration code is $(1.0, 1.0)^T$.

Problem 4

(a) First derivative formula

Show that

$$f'(t) \approx \frac{1}{2h}\left[-3f(t) + 4f(t+h) - f(t+2h)\right].$$

Determine the order of the truncation error.

Derivation:
By using the Taylor expansion for each point in the approximation,

$$\begin{aligned}
f(t+h) &= f(t) + f'(t)h + \frac{f''(t)h^2}{2} + \frac{f'''(t)h^3}{6} + \frac{f''''(t)h^4}{24} \\
f(t) &= f(t) \\
f(t+2h) &= f(t) + f'(t)2h + 2f''(t)h^2 + \frac{4f'''(t)h^3}{3} + \frac{2f''''(t)h^4}{3}
\end{aligned}$$

By substituting the expansions in the approximation,

$$\begin{aligned}
-3f(t) + 4f(t+h) &= f(t) + 4f'(t)h + 2f''(t)h^2 + \frac{2f'''(t)h^3}{3} + \frac{f''''(t)h^4}{6} \\
\frac{1}{2h}(-3f(t) + 4f(t+h)) - f(t+2h) &= \frac{1}{2h}(2f'(t)h - \frac{2h^3 f'''(\xi)}{3}) = f'(t) - \frac{h^2 f'''(\xi)}{3} \\
\frac{1}{2h}[-3f(t) + 4f(t+h) - f(t+2h)] - f'(t) &= -\frac{h^2 f'''(\xi)}{3}
\end{aligned}$$

The error term is of order $O(h^2)$.

(b) Second derivative formula

Show that

$$f''(t) \approx \frac{1}{12h^2}\left[-f(t+2h) + 16f(t+h) - 30f(t) + 16f(t-h) - f(t-2h)\right].$$

Determine the order of the truncation error.

Derivation:
Using the logic from the prior part, expanding all the points gives,

$$f(t+h) = f(t) + f'(t)h + \frac{f''(t)h^2}{2} + \frac{f'''(t)h^3}{6} + \frac{f''''(t)h^4}{24}$$
$$+ \frac{f^{(5)}(t)h^5}{120} + \frac{f^{(6)}(t)h^6}{720}$$
$$f(t) = f(t)$$
$$f(t+2h) = f(t) + f'(t)2h + 2f''(t)h^2 + \frac{4f'''(t)h^3}{3} + \frac{2f''''(t)h^4}{3}$$
$$+ \frac{4f^{(5)}(t)h^5}{15} + \frac{4f^{(6)}(t)h^6}{45}$$
$$f(t-h) = f(t) - f'(t)h + \frac{f''(t)h^2}{2} - \frac{f'''(t)h^3}{6} + \frac{f''''(t)h^4}{24}$$
$$- \frac{f^{(5)}(t)h^5}{120} + \frac{f^{(6)}(t)h^6}{720}$$
$$f(t-2h) = f(t) - f'(t)2h + 2f''(t)h^2 - \frac{4f'''(t)h^3}{3} + \frac{2f''''(t)h^4}{3}$$
$$- \frac{4f^{(5)}(t)h^5}{15} + \frac{4f^{(6)}(t)h^6}{45}$$

substituting into the approximations

$$-f(t+2h) + 16f(t+h) = 15f(t) + 14f'(t)h + 6f''(t)h^2 + \frac{4f'''(t)h^3}{3}$$
$$- \frac{2h^5 f^{(5)}(t)}{15} - \frac{h^6 f^{(6)}(t)}{15}$$
$$-30f(t) + 16f(t-h) = -14f(t) - 16f'(t)h + 8f''(t)h^2 - \frac{8f'''(t)h^3}{3} + \frac{2f''''(t)h^4}{3}$$
$$- \frac{2h^5 f^{(5)}(t)}{15} + \frac{h^6 f^{(6)}(t)}{45}$$
$$(-f(t+2h) + 16f(t+h)) + (-30f(t) + 16f(t-h)) = f(t) - 2f'(t)h + 14f''(t)h^2 - \frac{4f'''(t)h^3}{3} + \frac{2f''''(t)h^4}{3}$$
$$- \frac{4h^5 f^{(5)}(t)}{15}$$
$$((-f(t+2h) + 16f(t+h)) + (-30f(t) + 16f(t-h))) - f(t-2h) = 12f''(t)h^2 - \frac{6h^6 f^{(6)}(t)}{45}$$
$$\frac{1}{12h^2}(-f(t+2h) + 16f(t+h) - 30f(t) + 16f(t-h) - f(t-2h)) = f''(t) - \frac{h^4 f^{(6)}(t)}{90}$$
$$\frac{1}{12h^2}(-f(t+2h) + 16f(t+h) - 30f(t) + 16f(t-h) - f(t-2h)) - f''(t) = -\frac{h^4 f^{(6)}(t)}{90}$$

The error is of order $O(h^4)$.

Problem 5

Consider the two-point boundary-value problem

$$-\phi'' + (4x^2 + 2)\phi = 0, \qquad \phi(0) = 1, \qquad \phi(1) = e,$$

whose exact solution is

$$\phi(x) = e^{x^2}.$$

Write a program to solve the problem using the second-order finite-difference scheme. Use step sizes

$$h = 2^{-p}, \qquad p = 1, 2, \ldots, 14.$$

Solve the resulting linear system using:

1. LU factorization / Thomas algorithm;
2. Gradient descent;
3. Conjugate gradient.

For the iterative methods, state the initial guess and stopping criterion. Include your code, plots, and the requested error tables.

All iterative methods were ran with $x_0 = 0$ and stopping tolerance of 1×10^{-9}

Method 1: Thomas algorithm

Error Tables

Table 1: Errors using the Thomas algorithm.

h	$\|u_h - \phi_h\|_\infty$	$\|u_h - \phi_h\|_\infty / h$	$\|u_h - \phi_h\|_\infty / h^2$	$\|u_h - \phi_h\|_\infty / h^3$
5.0000×10^{-1}	6.8077×10^{-2}	1.3615×10^{-1}	2.7231×10^{-1}	5.4462×10^{-1}
2.5000×10^{-1}	2.0012×10^{-2}	8.0048×10^{-2}	3.2019×10^{-1}	1.2808
1.2500×10^{-1}	5.4792×10^{-3}	4.3834×10^{-2}	3.5067×10^{-1}	2.8054
6.2500×10^{-2}	1.3851×10^{-3}	2.2161×10^{-2}	3.5458×10^{-1}	5.6732
3.1250×10^{-2}	3.4860×10^{-4}	1.1155×10^{-2}	3.5697×10^{-1}	11.4230
1.5625×10^{-2}	8.7213×10^{-5}	5.5816×10^{-3}	3.5722×10^{-1}	22.8624
7.8125×10^{-3}	2.1807×10^{-5}	2.7913×10^{-3}	3.5729×10^{-1}	45.7330
3.9063×10^{-3}	5.4521×10^{-6}	1.3957×10^{-3}	3.5731×10^{-1}	91.4702
1.9531×10^{-3}	1.3630×10^{-6}	6.9787×10^{-4}	3.5731×10^{-1}	182.9429
9.7656×10^{-4}	3.4076×10^{-7}	3.4894×10^{-4}	3.5731×10^{-1}	365.8881
4.8828×10^{-4}	8.5188×10^{-8}	1.7447×10^{-4}	3.5731×10^{-1}	731.7615
2.4414×10^{-4}	2.1301×10^{-8}	8.7249×10^{-5}	3.5737×10^{-1}	1463.7983
1.2207×10^{-4}	5.3250×10^{-9}	4.3622×10^{-5}	3.5735×10^{-1}	2927.4504
6.1035×10^{-5}	4.1937×10^{-9}	6.8710×10^{-5}	1.1257	18444.1299

Table 2: Errors using the Conjugate Gradient method.

h	$\|u_h - \phi_h\|_\infty$	$\|u_h - \phi_h\|_\infty / h$	$\|u_h - \phi_h\|_\infty / h^2$	$\|u_h - \phi_h\|_\infty / h^3$
5.0000×10^{-1}	6.8077×10^{-2}	1.3615×10^{-1}	2.7231×10^{-1}	5.4462×10^{-1}
2.5000×10^{-1}	2.0012×10^{-2}	8.0048×10^{-2}	3.2019×10^{-1}	1.2808
1.2500×10^{-1}	5.4792×10^{-3}	4.3834×10^{-2}	3.5067×10^{-1}	2.8054
6.2500×10^{-2}	1.3851×10^{-3}	2.2161×10^{-2}	3.5458×10^{-1}	5.6732
3.1250×10^{-2}	3.4860×10^{-4}	1.1155×10^{-2}	3.5697×10^{-1}	11.4230
1.5625×10^{-2}	8.7213×10^{-5}	5.5816×10^{-3}	3.5722×10^{-1}	22.8624
7.8125×10^{-3}	2.1807×10^{-5}	2.7913×10^{-3}	3.5729×10^{-1}	45.7330
3.9063×10^{-3}	5.4521×10^{-6}	1.3957×10^{-3}	3.5731×10^{-1}	91.4702
1.9531×10^{-3}	1.3630×10^{-6}	6.9787×10^{-4}	3.5731×10^{-1}	182.9429
9.7656×10^{-4}	3.4076×10^{-7}	3.4894×10^{-4}	3.5731×10^{-1}	365.8881
4.8828×10^{-4}	8.5190×10^{-8}	1.7447×10^{-4}	3.5731×10^{-1}	731.7777
2.4414×10^{-4}	2.1297×10^{-8}	8.7233×10^{-5}	3.5731×10^{-1}	1463.5295
1.2207×10^{-4}	5.3239×10^{-9}	4.3613×10^{-5}	3.5728×10^{-1}	2926.8251
6.1035×10^{-5}	4.1742×10^{-9}	6.8390×10^{-5}	1.1205	18358.4111

Table 3: Errors using the Gradient Descent method.

h	$\|u_h - \phi_h\|_\infty$	$\|u_h - \phi_h\|_\infty / h$	$\|u_h - \phi_h\|_\infty / h^2$	$\|u_h - \phi_h\|_\infty / h^3$
5.0000×10^{-1}	6.8077×10^{-2}	1.3615×10^{-1}	2.7231×10^{-1}	5.4462×10^{-1}
2.5000×10^{-1}	2.0012×10^{-2}	8.0048×10^{-2}	3.2019×10^{-1}	1.2808
1.2500×10^{-1}	5.4247×10^{-3}	4.3398×10^{-2}	3.4718×10^{-1}	2.7775
6.2500×10^{-2}	1.3579×10^{-1}	2.1726	34.7624	556.1979
3.1250×10^{-2}	9.2697×10^{-1}	29.6631	949.2195	30375.0233
1.5625×10^{-2}	1.4913	95.4429	6108.3461	390934.1507
7.8125×10^{-3}	1.9168	245.3517	31405.0154	4019841.9717
3.9063×10^{-3}	2.2328	571.5924	146327.6659	37459882.4808
1.9531×10^{-3}	2.4386	1248.5529	639259.0594	327300638.4105
9.7656×10^{-4}	2.5624	2623.9028	2686876.4235	2751361457.6764
4.8828×10^{-4}	2.6333	5393.0959	11045060.4320	22620283764.6882
2.4414×10^{-4}	2.6727	10947.5173	44841030.7701	183668862034.2852
1.2207×10^{-4}	2.6941	22070.3813	180800563.6421	1481118217355.7407
6.1035×10^{-5}	2.7056	44328.4526	726277367.5724	11899328390305.924

Plots

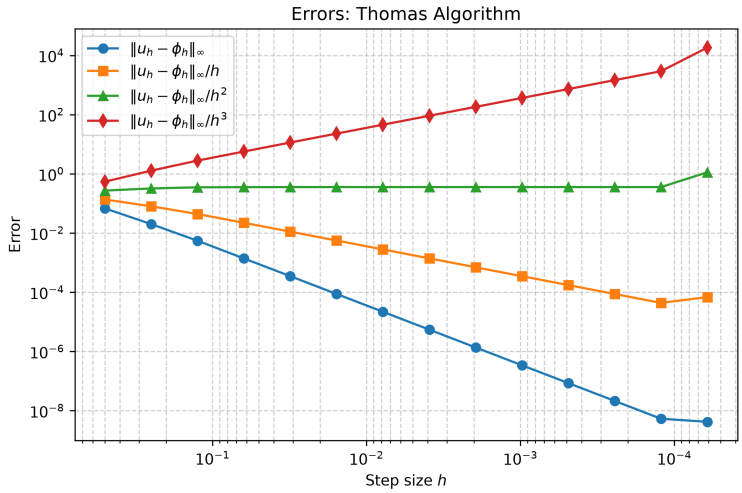


Figure 1: Error quantities for the Thomas algorithm.

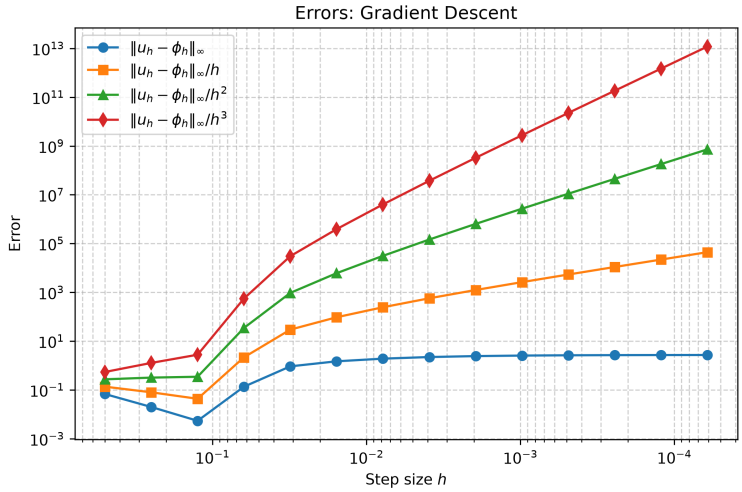


Figure 2: Error quantities for the Gradient Descent algorithm.

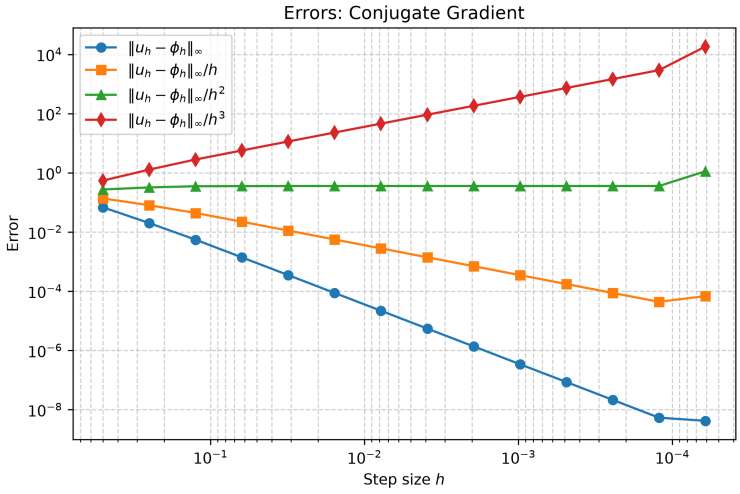


Figure 3: Error quantities for the Conjugate Gradient algorithm.